SIMT-Aware Lockstep Verification and Functional-Coverage Closure Methodology for an Open-Source RISC-V GPGPU: A Complete UVM 1.2 Environment

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Abstract

Open-source RISC-V GPGPUs ship without reference-model verification, functional-coverage models, or sign-off discipline, while concurrent fuzzing work demonstrates that real defects populate exactly this design class. This paper presents a complete UVM 1.2 verification environment and closure methodology for the Vortex RISC-V GPGPU, and the findings it produced on both sides of the comparison. The environment wraps a bus-master SIMT device with role-inverted agents, integrates the project's functional simulator as a per-configuration golden model over DPI-C, and renders verdicts through two checkers that are themselves qualified by permanent fault-injection guards: a bidirectional end-state scoreboard and a per-instruction, per-lane lockstep comparator governed by five alignment rules that we semi-

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formalize as precondition–soundness properties. A two-pass load-value feed makes fenceless multi-core programs instruction-granularity verifiable—verified modulo the fed race resolutions, with the assumption stated and, in the flagship case, discharged by a trace-level race witness. A three-layer coverage model adds what is, to our knowledge, the first published SIMT functional-coverage layer for RTL GPU verification (IPDOM divergence depth, scratchpad bank conflicts, coalescing classes), closed under machine-generated, RTL-cited exclusions enforced by a blocking waiver-integrity gate. The methodology surfaced externally corroborated RTL defects (a JALR LSB ISA deviation, a reset-relay X window behind a silenced assertion) and reference-model defects found by the lockstep itself. The checking stack is cheap to operate: lockstep adds no measurable wall-clock overhead at either end of the kernel-size spectrum and roughly 30% peak memory at scale. Stimulus *kind*, not volume, moves the coverage model: ninety additional scalar-generator programs closed zero bins, while fifty-seven axis-directed programs closed the SIMT targets.

Keywords: UVM, RISC-V, GPGPU, SIMT, functional verification, lockstep co-simulation, functional coverage, open-source hardware

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1. Introduction

Open-source silicon has an industrial verification problem with a sharpening edge. The RISC-V ecosystem maintains mature verification assets for scalar cores—step-and-compare environments, constrained-random generators, ISA coverage libraries [5, 7, 6, 9, 4, 10]—but its most complex open processor class, the GPGPU, has had none of this. Vortex [1, 2], the leading

open RISC-V GPGPU, ships a directed-kernel regression with no RTL-level reference checking and no functional-coverage model. Concurrently, FuzzGPU [3] demonstrated that a fuzzer can find real defects in exactly this class of design (20 previously unknown bugs, 10 CVEs, across Vortex and Ventus), converting the gap from an academic inconvenience into a security exposure.

This paper closes the other half of that problem: not bug discovery, but *coverage-quantified sign-off*. Its contributions:

1. a complete, configuration-parametric UVM 1.2 [11] environment for a bus-master SIMT GPGPU (Section 3);
2. per-instruction, per-lane lockstep for SIMT via five alignment rules, *semi-formalized* as precondition–soundness properties (Section 5);
3. a two-pass load-value feed that makes fenceless multi-core programs instruction-granularity verifiable, with its soundness assumption stated and, where possible, discharged by trace-level race witnesses (Section 6);
4. a three-layer coverage model including the first published SIMT functional-coverage layer for RTL GPU verification (Section 8);
5. an exclusion methodology—machine-generated, RTL-cited waivers under a blocking hits-invariant merge gate—elevated here to a named, reusable contribution (Section 9);
6. an evidence-classified findings register covering both the RTL and the reference model, with dispositions and external corroboration (Sections 10–10.2), and the verification-cost and per-generator marginal-coverage evidence that adoption requires (Section 11).

All honesty constraints of the conference version carry over verbatim (Section 12); every number in this paper traces to a banked artifact or is marked as pending.

2. Background and related work

2.1. The DUT and its reference models

Vortex organizes as clusters → sockets → cores → warps → threads with per-core caches, optional shared L2/L3, an immediate-post-dominator (IPDOM) reconvergence stack, and hardware barriers; the verified build is RV32IMAF at a pinned revision on QuestaSim. Vortex has no trap architecture—a property with verification consequences (Section 10). SimX, its functional simulator, is structurally an RVVI-shaped golden model [6]; Spike [8] cannot execute the SIMT instructions and serves only as a scalar, independent base-ISA cross-check (11,076-retirement three-way audit: zero mismatches, injection-proven non-vacuity). *The SIMT axis has no independent reference*—a structural ceiling we return to in the threats section.

2.2. Reference-model verification of processors

Step-and-compare is the standard of care for RISC-V CPUs. core-v-verif [5] established the open-source pattern with ImperasDV [9] over RVVI [6]; riscv-dv [7] supplies constrained-random programs with a trace-compare flow; DiffTest [4] industrialized per-instruction architectural comparison over DPI-C for XiangShan; OpenTitan [10] demonstrates sign-off-grade UVM on open silicon at SoC scale. All target scalar or SIMD-vector cores. This work extends the family along three axes: a bus-master DUT with role-inverted

agents, SIMT retirements (per-lane masked compare, split records, divergence), and weakly-ordered multi-core programs (the two-pass feed). Per-warp differential testing on this very DUT was concurrently demonstrated by FuzzGPU; our lockstep differs in the formalized alignment rules, the UVM packaging, and its role as one qualified checker inside a sign-off flow rather than a fuzzer's oracle.

2.3. GPU kernel verification and testing

Software-level kernel verification is mature: GPUVerify [15] proves race- and divergence-freedom of OpenCL/CUDA kernels; GKLEE [16] performs concolic kernel testing, explicitly targeting divergent warps, non-coalesced accesses, and shared-memory bank conflicts—the same three phenomena our coverage layer quantifies at the RTL level; WEFT [17] verifies producer–consumer synchronization in GPU programs. Learned program generation has precedent in compiler fuzzing [18]. These operate on kernel source, not RTL; our contribution begins where their remit ends.

2.4. RTL GPU testing and memory consistency

FuzzGPU [3] (USENIX Security 2026) is the first RTL GPU fuzzer: SIMT-aware generation, trace-driven differential testing against ISA simulators on Verilator models of Vortex and Ventus, 20 bugs and 10 CVEs. Its evaluation is structural-coverage-only, its campaigns time-boxed, and it carries no functional-coverage model, no checker qualification, and no sign-off discipline—by design. Our methodology is the complementary half, and the flows cross-validate: the JALR LSB deviation we report (R1) was independently found by their S2 (filed against the project's instruction-set simulator;

their RTL findings X1–X3, at a different pin, are disjoint from ours). Formal memory-consistency verification at RTL is addressed for CPU models by RTLCheck [21] and for GPU memory models in ASPLOS [22]; NVBitFI [20] brought systematic fault injection to GPUs for reliability studies. Concurrently, the Ventus project developed GVM, a UVM-like framework for its own design.

2.5. Checker qualification

Mutation-based qualification—proving checkers can fail—is established practice [19]; we apply the discipline end-to-end to a SIMT lockstep flow, which is what allows our pass rates and coverage to be read at face value.

3. The verification environment

[Ported text: role-inverted agents; DPI-C SimX; probes; end-state and lockstep checkers; assertion-aware verdicts; configurability; protocol assertions. Expanded treatment of the probe layer's sampling semantics and of the launch-space sequences.]

4. Verdicts and non-vacuity

[Ported text.]

5. Per-instruction lockstep for SIMT: five rules, semi-formalized

5.1. The alignment problem

Let D be the DUT's record stream per (core, warp), each record d carrying five fields (identifier, active-lane mask, PC, destination register, per-lane

values), and G the reference's instruction stream g over the same fields, in program order. Comparing D and G naively produces false mismatches from five structural divergences. Each rule below is stated as *precondition* $\Rightarrow$ *soundness property*: if the precondition holds for the DUT, the transformation cannot mask a genuine defect (it may only remove false positives), and the residual comparator is exact outside the declared exception classes.

5.2. Rule 1 — retire order (sort by issue counter)

Precondition: the per-(core, warp) issue counter is strictly monotone in program order and does not wrap within a run. *Property:* sorting D by `uuid` per (core, warp) yields program order; hence record d_i aligns with instruction g_i by position. A retirement whose (PC, rd) differ from g_i remains visible. (Observed DUT behavior satisfies the precondition: an earlier-issued instruction was observed retiring after two later ones; the arbiter is `VX_commit.sv:56-71`.)

5.3. Rule 2 — cross-key (program-order keying)

Precondition: each stream independently enumerates the same per-(core, warp) instruction sequence. *Property:* aligning by per-(core, warp) ordinal position is well-defined without any shared identifier; the DUT `uuid`'s high bits supply (core, warp) attribution with no RTL change. Model-side identifiers (SimX leaves `uuid` at zero) are never used as keys.

5.4. Rule 3 — split retirements (mask-union aggregation)

Precondition: every record of one architectural instruction shares its `uuid`, and the union of the records' thread masks equals the instruction's

active-lane set L (records may overlap; observed `0xd` then `0xe`). *Property:* aggregating records by `uuid` with mask union, then comparing per active lane, is equivalent to comparing the instruction itself: every lane in L is compared exactly once semantically, and no lane outside L contributes.

5.5. Rule 4 — load data (own tap, region-filtered)

Precondition: (i) a tap where true post-extension load values are observable (the commit arbiter's data field is stale for loads); (ii) a decidable predicate $V(a, v)$ that is true only when the two models' memory states agree at address a (initialized region, no preceding divergent write, reference value not initialization poison). *Property:* comparing only lanes with V is a sound *under*-approximation: it never masks a defect (non-V loads are deferred to the byte-exact end-state compare) at the cost of comparing fewer sites. (Empirically: 429 false mismatches on a known-good kernel without the filter; 74 compared / 113 filtered / 0 false with it.)

5.6. Rule 5 — volatile CSRs

Precondition: a statically known CSR class whose values are timing-defined rather than architecturally defined (`mcycle`/`minstret`/`mhpmcounter*`). *Property:* excluding their *data* (still comparing PC and destination) removes only comparisons with no architectural ground truth; soundness is preserved, completeness is reduced by exactly that class.

5.7. Declared exception classes and the composite claim

Beyond Rules 4–5, the comparator declares one bounded, logged tolerance: the DUT's `fsqrt.s` deviates by $\leq$ 1 ULP from correctly-rounded

IEEE 754 [14], tolerated only for `fsqrt` (same-sign, finite, $\leq$ 1 ULP via `fp32_within_ulp`), tallied in `n_fp_ulp_tol`, with a two-pass reconvergence feed re-checking all downstream operations bit-exactly. The composite soundness claim: *if the preconditions of Rules 1–3 hold and both models faithfully execute the same program, then any retirement whose (PC, destination, per-lane value) differs from the reference on an active lane, outside the declared exception classes, is reported as a mismatch.* The claim is conditional—the preconditions are DUT obligations an adopter must re-derive, listed as such in Section 12.

5.8. Validation

Lane-exact validation at five configurations (six programs) spanning a $2 \times 2 \times 3 \times 2$ axis space, including nested-divergence towers exercising Rule 3 through divergence and reconvergence; tallies and their provenance as in the conference version.

6. Verifying racy programs: the two-pass load-value feed

[Ported text with the assumption box and boundary discussion.]

7. Stimulus

[Ported text; simtgen axis status sentence updated after W4.]

8. A three-layer coverage model

[Ported text.]

9. Exclusion methodology: machine-generated, gate-checked waivers

[New text.]

10. Findings

10.1. RTL defects and hazards

[Ported text.]

10.2. Reference-model defects

[Ported text.]

11. Results

11.1. Closure campaign and bank provenance (W1)

The campaign concluded with a deliberate scope decision rather than a re-run for documentation-literalism: the divergence and memory closures are *in the suite bank* (`bank_1CL_1C_4W_4T_L2_20260909`: 23 covergroups, 504/524 bins = 96.18%, total 94.55% on the grown post-remediation coverage model); the barrier and VOTE/SHFL generator axes, built and verified by isolated merge (`simtgen_barrier_vote_shfl_20260910`), closed zero new bins because their targets (`cp_vote_shfl_op` 8/8, `cross_sfu_threads` 33/33) were already fully covered by dedicated directed kernels—an attribution correction to the initial reporting that we state rather than quietly keep. A relay-fixed 2CL bank (`bank_2CL_2C_4W_4T_relayfix_20260910`: 110 runs, 0 failures, 93.39% bins / 94.29% total) restores cross-configuration comparability on the fixed design. During this campaign the blocking hits-invariant waiver gate caught a real operator error—an exclusion set generated with 1CL keys

against a 2CL merge—and aborted the merge; this is the third class of defect the gate has caught in operation, recorded here as evidence for the exclusion methodology of Section 9.

11.2. Verification cost (W2)

Table 1: Wall-clock and peak-RSS cost of the checking stack (median-of-two reps; QuestaSim 2021.2, 1CL/1C/4W/4T). Four programs spanning the kernel-size spectrum: `vecadd_lite` (∼9.9k cycles), `diverge_lite` (divergence kernel), `fpu_test` (FPU), `wide_stress` (256 KB high-toggle).

Program	Mode	Wall clock	Peak RSS
vecadd_lite	plain	15.3 s	296 MB
vecadd_lite	lockstep	15.0 s	299 MB
vecadd_lite	lockstep+feed	14.8 s	299 MB
diverge_lite	plain	17.8 s	297 MB
diverge_lite	lockstep	18.2 s	301 MB
diverge_lite	lockstep+feed	18.1 s	301 MB
fpu_test	plain	21.0 s	296 MB
fpu_test	lockstep	20.7 s	301 MB†
fpu_test	lockstep+feed	21.0 s	301 MB
wide_stress	plain	851 s	549 MB
wide_stress	lockstep	871 s	708 MB
wide_stress	lockstep+feed	856 s	708 MB

Lockstep's wall-clock overhead is within run-to-run noise at every program size (±1.5 s on ∼15–21 s baselines; ±20 s—inside the inter-rep spread of plain mode—on the ∼860 s kernel). Peak memory shows one real step:

the second (SimX) model instance adds ∼29% RSS at the large working set and ∼1.5% at the small ones; the load-value feed adds no further measurable memory. The one FAIL in the grid is itself informative: `fpu\_test` under plain lockstep (feed disarmed) reproduces the known fsqrt 1-ULP deviation cascading into downstream mismatches—exactly the behavior Section 5's exception-class design predicts—while the same program under lockstep+feed passes with residual zero: the reconvergence feed, not the tolerance alone, absorbs the cascade. A measurement artifact is disclosed rather than hidden: an initial `wide_stress` run used an undersized timeout budget, produced false TIMEOUT verdicts on all six rows, and was discarded (budget artifact, OBS-042 precedent); all reported rows completed with rc=0.

11.3. FuzzGPU PoC reproduction at our pin (W3-B)

FuzzGPU's three Vortex RTL findings (filed at a different commit) were each exercised at our pin under this environment:

- **X1 (FPU compare-to-x0 unconditional writeback, PR #356): reproduced.** The PoC fires the RTL's own `invalid writeback register` runtime assertion at the PoC's exact `feq.s x0` PC; the defect is present and unfixed at our pin.

- **X2 (reserved M-extension `funct7` decode, PR #358): reproduced—and re-scoped.** The upstream fix lives entirely in the reference model's decoder; our RTL was always exact-match and never had the defect. A hand-encoded reserved opcode produces a genuine DUT-vs-SimX divergence in which the *hardware is right and the golden model is wrong* (DUT decodes ADD; SimX decodes MUL)— the paper's central thesis

(the reference model must be verified alongside the DUT) confirmed by an externally-sourced test case, and a caution against citing X2 as an RTL bug.

- **X3: static-only** (engineering cost of dynamic reproduction outweighed the value; documented, not attempted silently).

These runs are deliberately not suite members: X2's expected outcome is a real mismatch, so it is kept as a directed qualification case instead.

11.4. Marginal coverage per generator kind (W3-A)

The ordering—directed > simtgen > riscv-dv—quantifies, on one axis, the claim the seed farm first drew for riscv-dv alone: the SIMT stimulus gap needed a SIMT-aware generator, not more scalar seeds; and even a SIMT-aware generator cannot replace coverage-model validation (the bank-conflict and coalescing attribution lessons of Section 8).

11.5. Bug-discovery dynamics (W7)

Figure 1 plots cumulative register findings against project time, annotated with the methodology milestones that unlocked each cluster: the first lock-step bring-ups (reference-model defects became visible), the assertion-aware verdict gate (R10 surfaced the moment its silenced assertion was restored), the two-pass feed (racy-program findings), and the waiver gate (coverage-model defects). The curve is the paper's process argument in one picture: checking depth, added in stages, kept producing findings on both sides of the comparison for the entire project—there was no point at which the methodology saturated into vacuous passes.

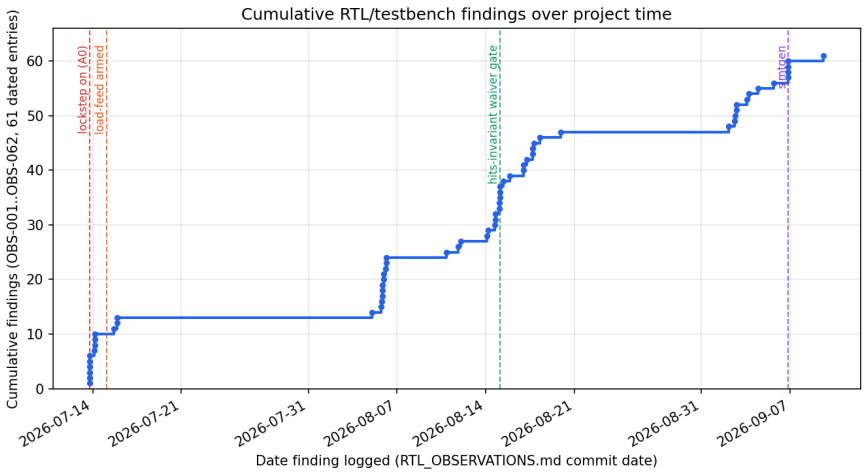

Figure 1: Cumulative findings (OBS register) over project time with methodology-milestone annotations. Data and reading: `figures/bug\_discovery\_curve.csv` / `.md` in the artifact.

12. Threats to validity

12.1. Internal validity

The two-pass feed's soundness rests on the assumption that a divergent in-region load is a genuine race resolution; we state it, discharge it with trace-level witnesses where possible, and classify rather than force-green elsewhere. Differential testing is structurally blind to stimulus faults (shared-binary blindness); one input class is closed by independent assertions, the class is reported open. Comparator tolerances are enumerated (load filter, volatile CSRs, fsqrt 1-ULP) and logged, never silent.

12.2. External validity

All results are one DUT at one pin with locally modified files (disclosed), one simulator, one build; the alignment rules' DUT preconditions are stated

so adopters can re-derive them; transfer beyond is belief, not demonstration. Open-source UVM simulation is maturing (upstream UVM 2017 elaborates in Verilator with constrained randomization), but SystemVerilog functional coverage remains unsupported there, so the coverage-driven half requires a commercial simulator today.

12.3. Conclusion validity

Coverage figures are per-configuration and never blended; frozen banks are never overwritten; the D-extension elaboration is scoped by inspection *and* confirmed by an EXT_D-off recompile with a coverpoint-by-coverpoint bin diff (zero scored bins change; the entire delta is confined to two weight-0 coverpoints already excluded from every reported percentage); per-mnemonic claims avoid aliased encodings; bank provenance and design state accompany every table.

13. Toward silicon sign-off

[Ported text.]

14. Conclusion

[Ported and updated once Section 11 is filled.]

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Table 2: Marginal covergroup-bin contribution per program kind. Directed kernels are hand-targeted at an identified gap, so their per-program rate is not comparable on the same axis as the two generators; the two generator rows are measured directly.

Generator	Programs	Marginal contribution
riscv-dv (seed farm, seeds 2–11 × 9 profiles)	90	**0 bins** (370/377 unchanged; toggle +0.06% only)
simtgen (divergence + memory axes)	57	`cp\_split\_depth` 3/4 → 4/4; `cp\_bank\_conflict` 0/3 → 3/3 (credit shared with the OBS-060 probe reclassification)
simtgen (barrier + vote_shfl axes)	6	**0 bins** (targets already covered by directed kernels `vote\_shfl`, `bar\_masks`, `sfu\_masks`)
Directed kernels (representative)	—	`lmem\_stress`: scratchpad toggle 57.5%→73.2%; `multicore\_isa`: closed the per-core gap across all 4 cores; `mshr\_flood`: 67,207 misses driven, target still unhit — itself finding OBS-031, not a closure